

House Price Prediction using XGBoost - Project Guide

Project Flow

- 1.Import Libraries
- 2.Load Dataset
- 3.Visualization (Histogram, Scatter, Heatmap)
- 4.Data Cleaning
- 5.Feature Engineering
- 6.Label Encoding
- 7.Train-Test Split
- 8.Feature Scaling
- 9.XGBoost Training
- 10.Model Evaluation
- 11.User Prediction

Data Cleaning

Remove duplicates.
Fill numerical missing values with median.
Fill categorical missing values with mode.
Remove unnecessary columns.

Feature Engineering

Create: price_per_sqm, house_age, total_rooms, luxury_house, area_category.

Model

Train XGBRegressor. Evaluate using R² Score and MAE. Save model with joblib.

Prediction

Take user inputs, generate engineered features, scale with StandardScaler, predict house price.